Pilot Feasibility Note

Day 2 Deliverable — first pass · validating Aitbaar on real UBL files before any live lending decision

1 OBJECTIVE

Validate — before Aitbaar makes or influences a single live lending decision — that it cuts turnaround and officer time **while holding or improving credit quality**, on real UBL SME files rather than the synthetic prototype dataset the model is trained on today. The pilot's job is to turn every assumed number in this note into a measured one.

2 PILOT SCOPE

- **Bank —** UBL. Fits on every criterion that matters: 11m customers, Temenos loan origination already live across five segments including SME, an existing WhatsApp banking channel, an IFC advisory engagement explicitly aimed at reducing SME loan turnaround time, and live SME products (Karobar Loan, PM Youth Business Loan at 5–7% markup).
- **Product —** one live SME product, Karobar Loan or PM Youth Business Loan (final choice with UBL SME Banking).
- **Geography —** one city, Lahore — consistent with the market conversations already held with a former bank loan-department staff member and a serving branch manager (Lahore, July 2026).
- **Site —** 1–2 branches, or one SME desk in a high-volume trading market. Retail & wholesale trade is ~45% of Pakistan's business establishments (Economic Census; see *data/DATA_CARD.md*) — the same segment our synthetic training data over-indexes on.
- **Volume —** ~500 applications.

3 APPROACH

Phased and strictly ordered — each phase must clear its own bar before the next begins. The officer makes the final decision on every file in every phase; this is a design principle carried from the working prototype (*POST /applications/{id}/decision*), not a pilot-only safeguard.

PHASE A · WEEKS 1–3
Shadow Mode
Aitbaar runs in parallel on real incoming files. No live decisions; the officer does not see the brief yet. Measures extraction accuracy and model behaviour on real documents before anyone relies on them.

PHASE B · WEEKS 4–8
Parallel Comparison
Officer sees the Aitbaar brief alongside the manual process; the manual decision stays authoritative. Measures turnaround, officer minutes, and officer/model agreement rate.

PHASE C · WEEKS 9–12
Limited Live Decisions
Human-in-the-loop live decisions, only once Phase A/B metrics clear an agreed threshold. Officer approves, declines, or requests exactly one document — as in the working prototype today.

4 SUCCESS METRICS

METRIC	BASELINE (ASSUMPTION)	TARGET WITH AITBAAR
Turnaround, application → decision	*2–4 weeks*	Same day – 48 hours
Officer time per file	*2–4 hours*	Under 10 minutes
Files per officer per day	*4–6*	Materially higher — quantified in Phase B
Branch visits per application	*3–5*	0 to apply, 1 to sign
First-time-right rate (zero re-requests)	*Not tracked today*	Baseline set in Phase A, then improved
Decision consistency across officers	*Not tracked; lives in one officer's head*	Structured, comparable, auditable
Model discrimination (holdout)	—	AUC 0.780 / KS 0.506 on synthetic data today; re-validated on real UBL outcomes during the pilot
Approval rate at a fixed expected default rate	*Unknown*	Established in Phase B, then held or improved
NPL vs. manual baseline	*Bank's own manual-process NPL*	Hold or improve vs. that baseline

Every figure in the "Baseline" column is a documented team assumption (Prototype Blueprint / Problem-Solution Fit Canvas), not a UBL-reported number. The pilot's first job — Phase A — is to replace each assumption with a measured baseline before any target is judged against it.

5 DATA & MODEL PLAN

TODAY

Model trained on 1,000 synthetic Pakistani SME records (*data/generate.py*, seed 42). Feature distributions and the ~20% base default rate are anchored to published SBP/SMEDA figures — SME NPL ~4.9% (Dec 2024) against a ~38% May-2023 stressed peak, retail & wholesale ~45% of establishments — not observed UBL data. Result: **AUC 0.780, KS 0.506** (*models/METRICS.md*), deliberately in the 0.78–0.85 band; a much higher score would signal the generator leaked the label into a feature, not genuine skill.

HONEST LIMITATION — STATED PLAINLY, NOT AROUND IT (DATA/DATA_CARD.MD)

These figures prove the pipeline works end-to-end; they are not evidence of real-world predictive power. Sample extraction documents used in development are mock-ups — extraction accuracy on real photographed CNICs and multi-bank statement formats is unproven. Six features (*registered, premises_owned, years_at_premises, account_age_months, has_existing_loan, existing_installment_pkr*) are not yet collected by the WhatsApp flow and default to dataset medians at inference — an explicit, documented approximation (*app/engine/scoring.py*), not a silent one.

PILOT PLAN

Phase A files build the first real-document extraction-accuracy measurement. Phase B decisions become labelled training data. Retrain on UBL's own portfolio outcomes before Phase C begins. Retraining is a parameter swap against the existing pipeline (*models/train.py*, *features.json*) — not an architecture change.

6 TECHNICAL FEASIBILITY

The working prototype exists end-to-end today — every line of the Prototype Blueprint's Basic Build Logic is marked Completed. The pilot integrates it, it does not rebuild it.

1 **Extract**	2 **Verify**	3 **Score**	4 **Explain**	5 **Brief**
Gemini vision reads CNIC, bank statement, utility bill into structured JSON	Deterministic rules cross-check fields, flag fraud — before scoring	XGBoost → repayment probability → risk tier A–D	SHAP attributes the score to its top 5 factors	LLM writes the rationale strictly from the SHAP output — never decides

- **Applicant side —** WhatsApp bot (whatsapp-web.js): full document checklist declared up front, explicit timestamped consent, accepts CNIC / bank statement / utility bill as photos or PDF plus a 5-question business questionnaire (*whatsapp-bot/src/flow.js*).
- **Verification detail —** CNIC format check, name-match between CNIC and bank account title via rapidfuzz at an 80% similarity threshold, income-consistency check at a 3x ratio, bounced-cheque flag at ≥3 — all run before the file reaches the model (*app/engine/verification.py*).
- **Officer side —** React/Vite dashboard (Vercel): application queue; one-page AI Credit Brief (score /100, risk tier, recommended limit, rationale, SHAP factors); Documents tab (extracted fields + verification flags per file); Audit Trail tab (every stage timestamped and actor-attributed); Approve / Reject / Request-exactly-one-document actions.
- **Storage —** Supabase Postgres (applications, scores, consent, audit log) + Supabase Storage (source documents). Backend on FastAPI/Railway.
- **Pilot integration points —** Temenos loan origination (mocked today via a stub API that shows exactly where it connects) and UBL's existing WhatsApp Business number (prototype currently runs on whatsapp-web.js) — both real integrations to complete in the pilot, not new builds.

7 REGULATORY & COMPLIANCE FEASIBILITY

- **Consent is a pipeline step, not a checkbox —** the applicant consents explicitly in the WhatsApp chat before any document is processed; consent text, timestamp and WhatsApp identifier are stored against the application.
- **Explainable by construction —** every score carries its SHAP factors; the brief LLM is limited to writing from those factors and cannot alter or invent a reason (*app/engine/rationale.py*) — the first question a credit-risk reviewer will ask.
- **Full audit trail —** every stage (created, submitted, extracted, verified, scored, decided) is logged with actor and timestamp; every officer decision is attributed to a named officer.
- **Human-in-the-loop, always —** the model only recommends (APPROVE / REVIEW / DECLINE via the policy layer — a HIGH-severity flag always forces REVIEW regardless of score); the officer alone submits the decision.
- **Policy alignment —** a direct answer to the SBP's call (Governor Jameel Ahmad, Pakistan Banking Summit 2026) for banks to use technology, data and fintech partnerships to reach the 750,000-borrower target.
- **Data protection —** no real customer data anywhere in the prototype today; pilot production data would sit inside UBL's own hosting environment, not the prototype's Supabase/Railway/Vercel stack. Team is tracking the Personal Data Protection Bill as it moves toward enactment.

8 RISKS & MITIGATIONS

RISK	MITIGATION
Extraction accuracy on real photographed documents is unproven (today: hand-made mock CNIC / statement / bill only)	Confidence-based verification flags surface to the officer, who reviews every file; the Pakistani-document edge-case library (angled photos, mixed Urdu/English bills, multi-bank statement formats) grows with every Phase A file.
Model bias in SME lending	Fairness check starts unusually clean: no proxy features like city or gender in the model's feature set (*app/engine/scoring.py*). Fairness checks are re-run explicitly after retraining on real UBL outcomes.
Data privacy / consent	Consent captured before processing, not after; timestamped and stored per application; production data hosted inside the bank's own environment.
Officer adoption	Officer-in-control design carried from the prototype — the brief recommends, the officer decides every file, matching what UBL staff asked for directly in team interviews ("simplify the officer's work," not remove him).
Synthetic-data limitation	Stated plainly in *DATA_CARD.md* rather than around it: today's AUC 0.780 proves the pipeline works end-to-end, not real-world predictive power. Phases A and B exist specifically to replace that assumption with measured, real-portfolio numbers before any live decision (Phase C).

9 COST & TIMELINE

PILOT COST

Small. Infra at pilot volume ~USD 50–150/month (Railway + Vercel + Supabase, same stack as the prototype); team time from the existing 4-engineer team; no incremental core-banking licensing cost beyond what UBL already pays for Temenos and its WhatsApp Business channel.

TIMELINE — 8–12 WEEKS

Weeks 1–3 Shadow Mode → Weeks 4–8 Parallel Comparison → Weeks 9–12 Limited Live Decisions (human-in-the-loop). See Approach, page 1.

GO / NO-GO CRITERIA — TO PROCEED TO FULL DEPLOYMENT

- Holdout AUC / KS on real UBL outcomes at or above the synthetic prototype's 0.780 / 0.506
- Turnaround measurably reduced vs. the Phase A manual baseline, with NPL held or improved
- Positive officer feedback on the brief and the queue — the throughput problem UBL staff raised directly, not model accuracy
- Compliance sign-off from UBL's risk and legal teams on consent, audit trail and explainability

"The AI recommends. The officer decides." Every pilot decision stays consent-based, explainable, and auditable. اعتبار